

Selection Models

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Motivation: Sample Selection Bias

- ▶ Ordinary Least Squares (OLS) assumes the sample is randomly drawn from the population.
- ▶ In many cases, we only observe the outcome variable y_i for a selected subset of individuals.
- ▶ Example: Wages are only observed for those who work.

$$y_i = w_i \text{ observed only if } s_i = 1$$

- ▶ If the decision to work is correlated with unobservables affecting wages, OLS is biased.

Outcome Equation (latent)

$$y_i^* = x_i' \beta + \varepsilon_i,$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

Selection Equation

$$s_i^* = z_i' \gamma + u_i,$$
$$s_i = 1[s_i^* > 0]$$
$$\begin{pmatrix} \varepsilon_i \\ u_i \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{pmatrix} \right)$$

Conditional Expectation and Bias

We only observe y_i when $s_i = 1$. Then:

$$\begin{aligned}\mathbb{E}[y_i \mid x_i, s_i = 1] &= x_i' \beta + \mathbb{E}[\varepsilon_i \mid u_i > -z_i' \gamma] \\ &= x_i' \beta + \rho \sigma \lambda(z_i' \gamma)\end{aligned}$$

where

$$\lambda(z_i' \gamma) = \frac{\phi(z_i' \gamma)}{\Phi(z_i' \gamma)}.$$

Thus:

$$\mathbb{E}[y_i \mid x_i, s_i = 1] = x_i' \beta + \rho \sigma \lambda(z_i' \gamma)$$

Heckman Two-Step Estimator

1. Step 1: Selection Equation (Probit)

$$s_i = 1[z_i'\gamma + u_i > 0].$$

Estimate $\hat{\gamma}$ via probit, compute

$$\hat{\lambda}_i = \frac{\phi(z_i'\hat{\gamma})}{\Phi(z_i'\hat{\gamma})}.$$

2. Step 2: Outcome Equation (Corrected OLS)

$$y_i = x_i'\beta + \rho\sigma\hat{\lambda}_i + \nu_i.$$

If $\rho \neq 0$, selection bias exists.

Note: standard errors must be corrected for the generated regressor $\hat{\lambda}_i$.

- ▶ ρ measures the correlation between the unobservables in the selection and outcome equations.
 - If $\rho \neq 0$, selection bias exists.
 - If $\rho = 0$, OLS on observed y_i is consistent.
- ▶ Identification relies on nonlinearity of $\lambda(z_i'\gamma)$, but it's better to have an **exclusion restriction**:

Some variables in z_i not in x_i .

Full Information Maximum Likelihood (FIML)

Joint likelihood for observed data:

$$\mathcal{L} = \prod_{i:s_i=1} f(y_i, s_i = 1 \mid x_i, z_i) \prod_{i:s_i=0} \Pr(s_i = 0 \mid z_i)$$

Assuming joint normality:

$$\log \mathcal{L} = \sum_{s_i=1} \log \left[\frac{1}{\sigma} \phi \left(\frac{y_i - x'_i \beta}{\sigma} \right) \Phi \left(\frac{z'_i \gamma + \rho(y_i - x'_i \beta)/\sigma}{\sqrt{1 - \rho^2}} \right) \right] + \sum_{s_i=0} \log [1 - \Phi(z'_i \gamma)]$$

Estimated via MLE; asymptotically efficient.

Empirical Example in R

```
n <- 2000

# regressors
educ <- rnorm(n, 12, 2)
exper <- rnorm(n, 10, 5)

# create correlated errors (epsilon for wage, u for selection) with rho != 0
rho <- 0.6                # correlation between wage error and selection error
sigma_eps <- 1
Sigma <- matrix(c(sigma_eps^2, rho*sigma_eps,
                  rho*sigma_eps, 1), nrow = 2)
errors <- mvrnorm(n, mu = c(0,0), Sigma = Sigma)
eps <- errors[,1]  # wage disturbances
u   <- errors[,2]  # selection disturbances

# -----
# Variant A: NO exclusion restriction
# selection and outcome share same regressors
# -----
s_star_A <- 0.5 + 0.3*educ - 0.2*exper + u
select_A <- as.integer(s_star_A > 0)

wage_star <- 2 + 0.10*educ + 0.05*exper + eps
wage_A <- ifelse(select_A==1, wage_star, NA)

ols_A <- lm(wage_A ~ educ + exper)
heck_A <- selection(select_A ~ educ + exper, wage_A ~ educ + exper, method = "ml")
```


Empirical Example in R: Heckman Results

```
Tobit 2 model (sample selection model)
Maximum Likelihood estimation
Newton-Raphson maximisation, 3 iterations
Return code 8: successive function values within relative tolerance limit (reltol)
Log-Likelihood: -2940.355
2000 observations (173 censored and 1827 observed)
8 free parameters (df = 1992)
Probit selection equation:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.19856    0.30870   0.643   0.52
educ         0.35155    0.03057  11.501 <2e-16 ***
exper       -0.21976    0.01460 -15.055 <2e-16 ***
Outcome equation:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.956873   0.153297  12.765 < 2e-16 ***
educ         0.095521   0.012536   7.619 3.91e-14 ***
exper        0.057020   0.005744   9.927 < 2e-16 ***
Error terms:
      Estimate Std. Error t value Pr(>|t|)
sigma  1.01697    0.01851  54.935 < 2e-16 ***
rho     0.54797    0.09150   5.989 2.5e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Empirical Example in R: Add Exclusion Restriction

Add excluded variable **kids** to selection equation:

```
kids <- rbinom(n, 1, 0.3)  # 0/1 indicator: has young children

s_star_B <- 0.5 + 0.3*educ - 0.2*exper - 0.8*kids + u  # kids reduces labor supply
select_B <- as.integer(s_star_B > 0)

wage_B <- ifelse(select_B==1, wage_star, NA)

ols_B <- lm(wage_B ~ educ + exper)
heck_B <- selection(select_B ~ educ + exper + kids, wage_B ~ educ + exper, method = "ml")
```

OLS vs. Heckman Selection Model (with exclusion restriction)

	OLS (observed)	Heckman (MLE)	Heckman (Exclusion)
educ	0.071*** (0.012)	0.096*** (0.013)	0.100*** (0.013)
exper	0.078*** (0.005)	0.057*** (0.006)	0.057*** (0.006)
Constant	2.154*** (0.150)	1.957*** (0.153)	1.898*** (0.156)
Observations	1,754	2,000	2,000
R ²	0.138		
Adjusted R ²	0.137		
Log Likelihood		−2,940.355	−2,903.073
ρ		0.548*** (0.091)	0.647*** (0.068)

Note:

*p<0.1; **p<0.05; ***p<0.01

► **Extensions:**

- Multinomial or dynamic selection.
- Semiparametric or nonparametric corrections.
- Panel data with selection.

► **Applications:**

- Wage equations (Heckman 1979)
- Labor force participation
- Credit approval models
- Censored health outcomes

Example: Borjas (1987)

- ▶ Consider two countries (0/1) (source and host).

$$\ln w_0 = \alpha_0 + u_0 \quad \text{with } u_0 \sim N(0, \sigma_0^2) \text{ source country}$$

$$\ln w_1 = \alpha_1 + u_1 \quad \text{with } u_1 \sim N(0, \sigma_1^2) \text{ host country}$$

- ▶ Now we allow for migration cost of C which he writes in hours: $\pi = \frac{C}{w_0}$.
- ▶ Assume workers know everything; you only see u_0 OR u_1 depending on country.
- ▶ Correlation in earnings is $\rho = \frac{\sigma_{01}}{\sigma_0 \sigma_1}$.

Example: Borjas (1987)

- ▶ Workers will migrate if:

$$(\alpha_1 - \alpha_0 - \pi) + (u_1 - u_0) > 0$$

- ▶ Who migrates? Probability of migration. Define $\nu = u_1 - u_0$.

$$\begin{aligned} P &= \Pr[\nu > (\alpha_0 - \alpha_1 + \pi)] = \Pr\left[\frac{\nu}{\sigma_\nu} > \frac{(\alpha_0 - \alpha_1 + \pi)}{\sigma_\nu}\right] \\ &= 1 - \Phi\left(\frac{(\alpha_0 - \alpha_1 + \pi)}{\sigma_\nu}\right) \equiv 1 - \Phi(z) \end{aligned}$$

- ▶ Higher $z \rightarrow$ less migration.

Example: Borjas (1987): How does selection work?

Construct **counterfactual wages** for workers in **source** country for those who immigrate:

- For now ignore mean differences $\alpha_0 = \alpha_1 = \alpha$.

$$\begin{aligned}\mathbb{E}(w_0 | \text{Immigrate}) &= \alpha + \mathbb{E}\left(u_0 \mid \frac{\nu}{\sigma_\nu} > z\right) \\ &= \alpha + \sigma_0 \cdot \mathbb{E}\left(\frac{u_0}{\sigma_0} \mid \frac{\nu}{\sigma_\nu} > z\right)\end{aligned}$$

- Wages depend on:

1. Mean earnings in the source country
2. Both error terms (u_0, u_1) through ν
3. Implicitly, it also depends on the correlation between the error terms.

Example: Borjas (1987): How does selection work?

- If everything is normal, we just run univariate regression $\mathbb{E}(u_0|\nu) = \frac{\sigma_{0\nu}}{\sigma_\nu^2}\nu$:

$$\mathbb{E}\left(\frac{u_0}{\sigma_0} \middle| \frac{\nu}{\sigma_\nu}\right) = \frac{1}{\sigma_0} \cdot \frac{\sigma_{0\nu}}{\sigma_\nu^2} \cdot \frac{\sigma_\nu^2}{\sigma_\nu^2} \cdot \nu = \frac{\sigma_{0\nu}}{\sigma_0\sigma_\nu} \frac{\nu}{\sigma_\nu} = \rho_{0\nu} \frac{\nu}{\sigma_\nu}$$

$$\begin{aligned}\mathbb{E}(w_0 | \text{Immigrate}) &= \alpha_0 + \sigma_0 \cdot \mathbb{E}\left(\frac{u_0}{\sigma_0} \middle| \frac{\nu}{\sigma_\nu} > z\right) \\ &= \alpha_0 + \rho_{0\nu} \cdot \sigma_0 \cdot \mathbb{E}\left(\frac{\nu}{\sigma_\nu} \middle| \frac{\nu}{\sigma_\nu} > z\right) \\ &= \alpha_0 + \rho_{0\nu} \cdot \sigma_0 \left(\frac{\phi(z)}{1 - \Phi(z)}\right)\end{aligned}$$

- This hazard rate of the standard normal has a special name **Inverse Mills Ratio** $\mathbb{E}[x|x > z]$.

Example: Borjas (1987): How does selection work?

- ▶ A similar expression for those who do immigrate:

$$\begin{aligned}\mathbb{E}(w_1 | \text{Immigrate}) &= \alpha_1 + \mathbb{E}\left(u_1 \mid \frac{\nu}{\sigma_\nu} > z\right) \\ &= \alpha_1 + \rho_{1\nu}\sigma_1 \left(\frac{\phi(z)}{\Phi(-z)}\right)\end{aligned}$$

- ▶ We can re-write both expressions in terms of the **Inverse Mills Ratio**

$$\begin{aligned}\mathbb{E}(w_0 | \text{Immigrate}) &= \alpha_0 + \rho_{0\nu} \sigma_0 \left(\frac{\phi(z)}{1 - \Phi(z)} \right) \\ &= \alpha_0 + \frac{\sigma_0 \sigma_1}{\sigma_\nu} \left(\rho - \frac{\sigma_0}{\sigma_1} \right) \left(\frac{\phi(z)}{1 - \Phi(z)} \right) \\ \mathbb{E}(w_1 | \text{Immigrate}) &= \alpha_1 + \rho_{1\nu} \sigma_1 \left(\frac{\phi(z)}{1 - \Phi(z)} \right) \\ &= \alpha_1 + \frac{\sigma_0 \sigma_1}{\sigma_\nu} \left(\frac{\sigma_1}{\sigma_0} - \rho \right) \left(\frac{\phi(z)}{1 - \Phi(z)} \right)\end{aligned}$$

Where $\rho = \sigma_{01} / \sigma_0 \sigma_1$.

Let $Q_0 = \mathbb{E}(u_0|I = 1)$, $Q_1 = \mathbb{E}(u_1|I = 1)$ (expected **skill** of immigrants).

- ▶ Immigrants are positively selected and above average (Q_0, Q_1) > 0 and $\frac{\sigma_1}{\sigma_0} > 1$ and $\rho > \frac{\sigma_0}{\sigma_1}$
 - $\frac{\sigma_1}{\sigma_0} > 1$ returns to “skill” are higher in host country.
 - $\rho > \frac{\sigma_0}{\sigma_1}$ correlation between valued skills in both counties is high (similar skills valued in both countries).
- ▶ Best and brightest leave because returns to skill are too low in home country.

We swap the standard deviations:

- ▶ Immigrants are negatively selected and below average ($Q_0, Q_1) < 0$ and $\frac{\sigma_1}{\sigma_0} > 1$ and $\rho > \frac{\sigma_0}{\sigma_1}$
 - $\frac{\sigma_0}{\sigma_1} > 1$ returns to “skill” are lower in host country.
 - $\rho > \frac{\sigma_1}{\sigma_0}$ correlation between valued skills in both counties is high (similar skills valued in both countries).
- ▶ Compressed wage structure attracts the low skill types because it provides “insurance” or “subsidizes” low wage workers.

Refugee/Superman Sorting?

- ▶ Immigrants are below average at home and above average in host ($Q_0 < 0, Q_1 > 1$) and $\frac{\sigma_1}{\sigma_0} > 1$:
 - $\rho < \min\left(\frac{\sigma_1}{\sigma_0}, \frac{\sigma_0}{\sigma_1}\right)$ being below average in source country makes you above average in host country.
- ▶ You are a nerdy intellectual in a country that values physical labor, or are otherwise discriminated against in the labor market.

The missing (fourth) case:

- ▶ Mathematically impossible $\rho > \max\left(\frac{\sigma_1}{\sigma_0}, \frac{\sigma_0}{\sigma_1}\right)$

What can we learn here?

- ▶ Heckman won a Nobel Prize for his work on selection...
- ▶ You need to know what an **inverse Mills ratio** is
- ▶ But today it is hard to get away with strong parametric assumptions (bivariate normal) on error terms.
- ▶ Doing MLE with a fully normal model is not a terrible place to start sometimes
 - Sometimes helpful to know how bad the selection problem might be.
 - But you probably need **exclusion restrictions**!
- ▶ R package is `sampleSelection` and see <https://rpubs.com/hacamvan/316839> and <https://cran.r-project.org/web/packages/sampleSelection/vignettes/selection.pdf>.